## Indications And Usage

INDICATIONS AND USAGE Pindolol tablets are indicated in the management of hypertension. It may be used alone or concomitantly with other antihypertensive agents, particularly with a thiazide-type diuretic.

## Dosage And Administration

DOSAGE AND ADMINISTRATION The dosage of pindolol tablets should be individualized. The recommended initial dose of pindolol tablets is 5 mg b.i.d. alone or in combination with other antihypertensive agents. An antihypertensive response usually occurs within the first week of treatment. Maximal response, however, may take as long as or occasionally longer than 2 weeks. If a satisfactory reduction in blood pressure does not occur within 3 to 4 weeks, the dose may be adjusted in increments of 10 mg/day at these intervals up to a maximum of 60 mg/day.

## Contraindications

CONTRAINDICATIONS Pindolol tablets are contraindicated in: 1) bronchial asthma; 2) overt cardiac failure; 3) cardiogenic shock; 4) second and third degree heart block; 5) severe bradycardia. (See WARNINGS .)

## Warnings

WARNINGS Cardiac Failure Sympathetic stimulation may be a vital component supporting circulatory function in patients with congestive heart failure, and its inhibition by beta-blockade may precipitate more severe failure. Although beta-blockers should be avoided in overt congestive heart failure, if necessary, pindolol can be used with caution in patients with a history of failure who are well-compensated, usually with digitalis and diuretics. Both digitalis glycosides and beta-blockers slow atrioventricular conduction and decrease heart rate. Concomitant use can increase risk of bradycardia. Beta-adrenergic blocking agents do not abolish the inotropic action of digitalis on heart muscle. In Patients Without A History of Cardiac Failure In patients with latent cardiac insufficiency, continued depression of the myocardium with beta-blocking agents over a period of time can in some cases lead to cardiac failure. At the first sign or symptom of impending cardiac failure, patients should be fully digitalized and/or be given a diuretic, and the response observed closely. If cardiac failure continues, despite adequate digitalization and diuretic, pindolol therapy should be withdrawn (gradually, if possible). Exacerbation of Ischemic Heart Disease Following Abrupt Withdrawal Hypersensitivity to catecholamines has been observed in patients withdrawn from beta-blocker therapy; exacerbation of angina and, in some cases, myocardial infarction have occurred after abrupt discontinuation of such therapy. When discontinuing chronically administered pindolol, particularly in patients with ischemic heart disease, the dosage should be gradually reduced over a period of 1 to 2 weeks and the patient should be carefully monitored. If angina markedly worsens or acute coronary insufficiency develops, pindolol administration should be reinstituted promptly, at least temporarily, and other measures appropriate for the management of unstable angina should be taken. Patients should be warned against interruption or discontinuation of therapy without the physician's advice. Because coronary artery disease is common and may be unrecognized, it may be prudent not to discontinue pindolol therapy abruptly even in patients treated only for hypertension. Nonallergic Bronchospasm (e.g., chronic bronchitis, emphysema) - Patients with Bronchospastic Diseases Should in General Not Receive Beta-Blockers Pindolol should be administered with caution since it may block bronchodilation produced by endogenous or exogenous catecholamine stimulation of beta 2 receptors. Major Surgery Because beta-blockade impairs the ability of the heart to respond to reflex stimuli and may increase the risks of general anesthesia and surgical procedures, resulting in protracted hypotension or low cardiac output, it has generally been suggested that such therapy should be gradually withdrawn several days prior to surgery. Recognition of the increased sensitivity to catecholamines of patients recently withdrawn from beta-blocker therapy, however, has made this recommendation controversial. If possible, beta-blockers should be withdrawn well before surgery takes place. In the event of emergency surgery, the anesthesiologist should be informed

that the patient is on beta-blocker therapy. The effects of pindolol can be reversed by administration of beta-receptor agonists such as isoproterenol, dopamine, dobutamine, or norepinephrine. Difficulty in restarting and maintaining the heart beat has also been reported with beta-adrenergic receptor blocking agents. Diabetes and Hypoglycemia Beta-adrenergic blockade may prevent the appearance of premonitory signs and symptoms (e.g., tachycardia and blood pressure changes) of acute hypoglycemia. This is especially important with labile diabetics. Beta-blockade also reduces the release of insulin in response to hyperglycemia; therefore, it may be necessary to adjust the dose of antidiabetic drugs. Thyrotoxicosis Beta-adrenergic blockade may mask certain clinical signs (e.g., tachycardia) of hyperthyroidism. Patients suspected of developing thyrotoxicosis should be managed carefully to avoid abrupt withdrawal of beta-blockade which might precipitate a thyroid crisis.

## Precautions

PRECAUTIONS Impaired Renal or Hepatic Function Beta-blocking agents should be used with caution in patients with impaired hepatic or renal function. Poor renal function has only minor effects on pindolol clearance, but poor hepatic function may cause blood levels of pindolol to increase substantially. Information for Patients Patients, especially those with evidence of coronary artery insufficiency, should be warned against interruption or discontinuation of pindolol therapy without the physician's advice. Although cardiac failure rarely occurs in properly selected patients, patients being treated with beta-adrenergic blocking agents should be advised to consult the physician at the first sign or symptom of impending failure. Drug Interactions Catecholamine-depleting drugs (e.g., reserpine) may have an additive effect when given with beta-blocking agents. Patients receiving pindolol plus a catecholamine-depleting agent should, therefore, be closely observed for evidence of hypotension and/or marked bradycardia which may produce vertigo, syncope, or postural hypotension. Pindolol has been used with a variety of antihypertensive agents, including hydrochlorothiazide, hydralazine, and guanethidine without unexpected adverse interactions. Pindolol has been shown to increase serum thioridazine levels when both drugs are coadministered. Pindolol levels may also be increased with this combination. Risk of Anaphylactic Reaction While taking beta blockers, patients with a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated challenge, either accidental, diagnostic, or therapeutic. Such patients may be unresponsive to the usual doses of epinephrine used to treat allergic reactions. Carcinogenesis, Mutagenesis, Impairment of Fertility In chronic oral toxicologic studies (1 to 2 years) in mice, rats, and dogs, pindolol did not produce any significant toxic effects. In 2-year oral carcinogenicity studies in rats and mice in doses as high as 59 mg/kg/day and 124 mg/kg/day (50 and 100 times the maximum recommended human dose), respectively, pindolol did not produce any neoplastic, preneoplastic, or nonneoplastic pathologic lesions. In fertility and general reproductive performance studies in rats, pindolol caused no adverse effects at a dose of 10 mg/kg. In the male fertility and general reproductive performance test in rats, definite toxicity characterized by mortality and decreased weight gain was observed in the group given 100 mg/kg/day. At 30 mg/kg/day, decreased mating was associated with testicular atrophy and/or decreased spermatogenesis. This response is not clearly drug-related, however, as there was no dose-response relationship within this experiment and no similar effect on testes of rats administered pindolol as a dietary admixture for 104 weeks. There appeared to be an increase in prenatal mortality in males given 100 mg/kg but development of offspring was not impaired. In females administered pindolol prior to mating through day 21 of lactation, mating behavior was decreased at 100 mg/kg and 30 mg/kg. At these dosages there also was increased mortality of offspring. Prenatal mortality was increased at 10 mg/kg but there was not a clear dose-response relationship in this experiment. There was an increased resorption rate at 100 mg/kg observed in females necropsied on the 15th day of gestation. Pregnancy Teratogenic Effects Category B Studies in rats and rabbits exceeding 100 times the maximum recommended human doses, revealed no embryotoxicity or teratogenicity. Since there are no adequate and well controlled studies in pregnant women, and since animal reproduction studies are not always predictive of human response, pindolol, as with any drug, should be employed during pregnancy only if the potential benefit justifies the potential risk to the fetus. Nursing Mothers Since pindolol is secreted in human milk, nursing should not be undertaken by mothers receiving the drug. Pediatric Use Safety and effectiveness in pediatric patients have not been established.

## Adverse Reactions

ADVERSE REACTIONS Most adverse reactions have been mild. The incidences listed in the following table are derived from 12-week comparative double-blind, parallel design trials in hypertensive patients given pindolol as monotherapy, given various active control drugs as monotherapy, or given placebo. Data for pindolol and the positive controls were pooled from several trials because no striking differences were seen in the individual studies, with one exception. When considering all adverse reactions reported, the frequency of edema was noticeably higher in positive control trials (16% pindolol vs. 9% positive control) than in placebo controlled trials (6% pindolol vs. 3% placebo). The table includes adverse reactions either volunteered or elicited, and at least possibly drug-related, which were reported in greater than 2% of pindolol patients and other selected important reactions. ADVERSE REACTIONS WHICH WERE VOLUNTEERED OR ELICITED (and at least possibly drug-related) Body System/ Adverse Reactions Pindolol (N=322)% Active Controls Active Controls: Patients received either propranolol, α-methyldopa or a diuretic (hydrochlorothiazide or chlorthalidone). (N=188)% Placebo (N=78)% Central Nervous System Bizarre or Many Dreams 5 0 6 Dizziness 9 11 1 Fatigue 8 4 4 Hallucinations <1 0 0 Insomnia 10 3 10 Nervousness 7 3 5 Weakness 4 2 1 Autonomic Nervous System Paresthesia 3 1 6 Cardiovascular Dyspnea 5 4 6 Edema 6 3 1 Heart Failure <1 <1 0 Palpitations <1 1 0 Musculoskeletal Chest Pain 3 1 3 Joint Pain 7 4 4 Muscle Cramps 3 1 0 Muscle Pain 10 9 8 Gastrointestinal Abdominal Discomfort 4 4 5 Nausea 5 2 1 Skin Pruritus 1 <1 0 Rash <1 <1 1 The following selected (potentially important) adverse reactions were seen in 2% or fewer patients and their relationship to pindolol is uncertain. CENTRAL NERVOUS SYSTEM: anxiety, lethargy; AUTONOMIC NERVOUS SYSTEM: visual disturbances, hyperhidrosis; CARDIOVASCULAR: bradycardia, claudication, cold extremities, heart block, hypotension, syncope, tachycardia, weight gain; GASTROINTESTINAL: diarrhea, vomiting; RESPIRATORY: wheezing; UROGENITAL: impotence, pollakiuria; MISCELLANEOUS: eye discomfort or burning eyes.

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## Pregnancy

Pregnancy Teratogenic Effects Category B Studies in rats and rabbits exceeding 100 times the maximum recommended human doses, revealed no embryotoxicity or teratogenicity. Since there are no adequate and well controlled studies in pregnant women, and since animal reproduction studies are not always predictive of human response, pindolol, as with any drug, should be employed during pregnancy only if the potential benefit justifies the potential risk to the fetus.

## Nursing Mothers

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## Pediatric Use

Pediatric Use Safety and effectiveness in pediatric patients have not been established.

## Overdosage

OVERDOSAGE No specific information on emergency treatment of overdosage is available. Therefore, on the basis of the pharmacologic actions of pindolol, the following general measures should be employed as appropriate in addition to gastric lavage: Excessive Bradycardia: administer atropine; if there is no response to vagal blockade, administer isoproterenol cautiously. Cardiac Failure: digitalize the patient and/or administer diuretic. It has been reported that glucagon may be useful in this situation. Hypotension: administer vasopressors, e.g., epinephrine or norepinephrine, with serial monitoring of blood pressure. (There is evidence that epinephrine may be the drug of choice.) Bronchospasm: administer a beta 2 stimulating agent such as isoproterenol and/or a theophylline derivative. A case of an acute overdosage has been reported with an intake of 500 mg of pindolol by a hypertensive patient. Blood pressure increased and heart rate was 80 beats/min. Recovery was uneventful. In another case, 250 mg of pindolol was taken with 150 mg diazepam and 50 mg nitrazepam, producing coma and hypotension. The patient recovered in 24 hours.

## Description

DESCRIPTION Pindolol, a synthetic beta-adrenergic receptor blocking agent with intrinsic sympathomimetic activity is 1-(Indol-4-yloxy)-3-(isopropylamino)-2-propanol. C 14 H 20 N 2 O 2 M.W. 248.32 Pindolol, USP is a white to off-white, crystalline powder having a faint odor which is practically insoluble in water; slightly soluble in methanol; and very slightly soluble in chloroform. Each tablet for oral administration contains pindolol and the following inactive ingredients: corn starch, lactose monohydrate, pregelatinized corn starch, colloidal silicon dioxide, magnesium stearate, sodium starch glycolate and microcrystalline cellulose.

## Clinical Pharmacology

CLINICAL PHARMACOLOGY Pindolol is a nonselective beta-adrenergic antagonist (beta-blocker) which possesses intrinsic sympathomimetic activity (ISA) in therapeutic dosage ranges but does not possess quinidine-like membrane stabilizing activity.

## Information For Patients

Information for Patients Patients, especially those with evidence of coronary artery insufficiency, should be warned against interruption or discontinuation of pindolol therapy without the physician's advice. Although cardiac failure rarely occurs in properly selected patients, patients being treated with beta-adrenergic blocking agents should be advised to consult the physician at the first sign or symptom of impending failure.